1 1 $z$ $AB$

$AB$ $C$ $C'$ $i$

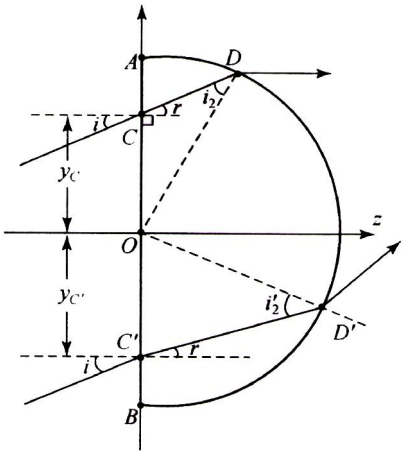
$r$

$D$ $D'$

$i_2$ $i'_2$ $\triangle OCD$ $O$ $C$

$y_c$

$$\frac{y_c}{\sin i_2} = \frac{R}{\sin(90° + r)}, \qquad y_c = \frac{\sin i_2}{\cos r} R \ . \qquad (1)$$



1

$\triangle OC'D'$ $O$ $C'$

$$\frac{y_{c'}}{\sin i'_2} = \frac{R}{\sin(90° - r)}, \qquad y_{c'} = \frac{\sin i'_2}{\cos r} R \ . \qquad (2)$$

$i$ $r$ $i_2$ $i'_2$

$$i_{20} = \arcsin(1/n) \approx 41.8° \qquad (3)$$

$i_2 = i'_2 = i_{20}$ (1), ( 2)

$$y_{oc} = y_{oc'} = \frac{\sin i_{20}}{\cos r} R \ , \qquad (4)$$

$$d = 2y_{oc} = 2\frac{\sin i_{20}}{\cos r} R \ . \qquad (5)$$

$y_c > y_{oc}$ $y_{c'} > y_{oc'}$

$i_{20}$ $r$ $i$ (4)

$r$ 0 $i$ 0 $d$

$$d_{min} = 2R\sin i_{20} = 1.33\,R \ . \qquad (6)$$

$i \to 90°$ $r \to 41.8°$ $r = 41.8°$

(4) $d_{max}$ 1.79 $R$ . ( 7)

2 2 $\varphi$ $Oz$ $OD$

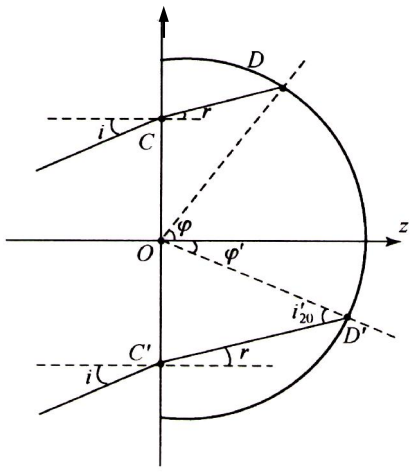
$\varphi'$ $Oz$ $OD'$

$$\varphi = i_{20} + r \ , \qquad (8)$$

$$\varphi' = i_{20} - r \ , \qquad (9)$$

$$\theta = \varphi + \varphi' = 2i_{20} \approx 83.6° \ . \qquad (10)$$

$\theta$ $i$ $\theta$ $i$



2